2 Trolley A and trolley B are on a horizontal, frictionless bench. Trolley A moves to the right with a constant velocity $u = 0.44\,\text{m/s}$. Trolley B is stationary.

Fig. 2.1 shows trolley A before it collides with trolley B.

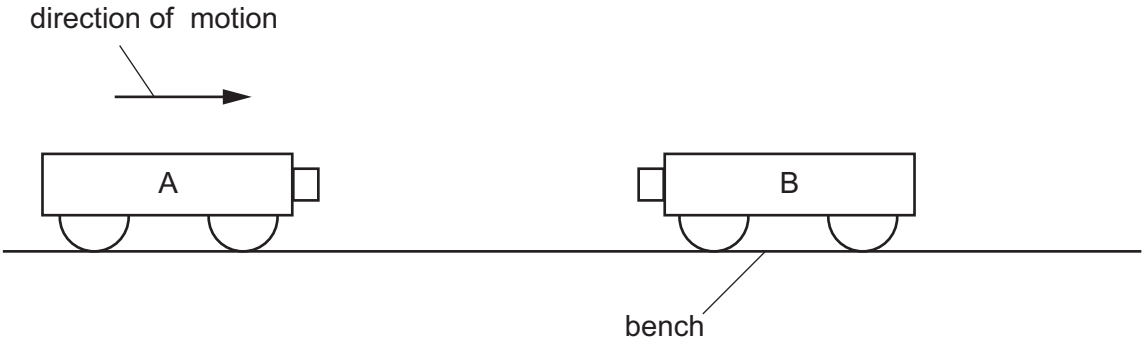

Fig. 2.1 (not drawn to scale)

(a) State the momentum of trolley B before the collision. Explain your answer.

statement .....

explanation .....

.....

[1]

(b) After the collision, the two trolleys are joined together and travel with a constant velocity $v = 0.18\,\text{m/s}$ to the right. The mass of trolley A is 0.75 kg.

Calculate the mass of trolley B.

mass of trolley B = ... [3]

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(c) (i) The trolleys move onto a rough surface which exerts a constant force F on the trolleys and brings them to rest in 2.6 s.

Calculate F.

F = ... [2]

(ii) A different rough surface exerts a smaller resistive force on the trolleys. State how this affects the time taken to bring the trolleys to rest. Explain your answer.

statement .....

explanation .....

.....

[1]

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